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YOLO4SWD: an improved YOLOv11-based method for waste detection and classification

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To make sorting more efficient, promote the recycling of resources, and lower the costs of environmental management, it is important to be able to accurately identify waste categories. However, current waste detection methods encounter substantial obstacles, such as inadequate feature extraction for diminutive waste objects, disproportionate contextual information application in intricate backgrounds, and a notable decline in classification accuracy in occlusion situations. To address these limitations, this study introduces an enhanced waste detection model derived from YOLOv11n, termed YOLO4SWD (YOLO for Smart Waste Detection), which integrates three principal innovations. (1)The CoordA module is embedded into the backbone to accurately capture fine local features of tiny waste and boost feature discrimination between waste and low-contrast backgrounds. (2) A lightweight BasicRFB block is integrated into the neck layer to enlarge the model's effective receptive field.(3) The FSEAM module is adopted in the detection head to preserve the effective features of unoccluded regions and block interference from occluded regions, thereby improving the accuracy of waste detection in scenarios where occlusions are common. The experimental results show that the proposed YOLO4SWD model is much better than the baseline YOLOv11n. The proposed model exhibits robust detection performance, achieving 81.6%, 69.9%, and 62.5% in mAP@0.5, mAP@0.75, and mAP@0.5:0.95, respectively. This indicates that it performs better in detecting trash in complex environments.

Keywords Smart Waste Detection, YOLO, CoordA module, BasicRFB module, FSEAM module

As economies grow and living standards rise, it becomes unavoidable that people will produce a lot of municipal solid waste every day. Studies indicate that the amount of waste produced around the world is still going up and is expected to reach 46 billion tons by 2050[1]. It is becoming more and more clear how important it is to manage household waste. It is clear that such action will help reduce pollution and the spread of disease. Incorrect ways of handling household waste can have very negative effects on the environment in a region, making life difficult for residents. Even though most people know that waste is bad for the environment and people's health, both developing and developed countries still have a lot of work to do when it comes to managing it in their cities and public spaces.

In the field of waste detection, there has long been a lack of standardized benchmarks and dedicated datasets. Recently, this research gap has been gradually filled through the construction of new databases and comprehensive review studies, providing crucial support for improving recognition accuracy and classification efficiency. Object detection has become an important research direction in the field of computer vision[2]. Liu et al. proposed EcoDetect-YOLO, a garbage exposure detection algorithm based on the YOLOv5s framework, which demonstrates stronger deep garbage detection capabilities in complex urban environments, improving the accuracy and speed of small object detection[3]. Li et al. put forward a multi-model cascaded CNN that uses SSD, YOLOv4, and Faster R-CNN to cut down on wrong predictions when finding household waste[4]. Malik et al. applied image recognition technologies to revolutionize urban solid waste management, promoting automated sorting and recycling toward sustainable development[5]. Mao et al. employed an optimized convolutional neural network architecture combined with genetic algorithms to deploy a waste classification model, achieving higher accuracy using datasets such as TrashNet[6]. Guo et al. integrated CNN into an advanced stereovision system, enabling accurate and efficient multi-object garbage detection for road cleaning robots, thereby improving efficiency and accuracy[7]. Li et al. further improved the YOLOv8s object detection model, introducing a lightweight yet high-performance waste detection framework that provides an efficient automated solution

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for garbage detection[8]. Wang et al. proposed a lightweight waste detection model, namely YOLO-WasNet, to mitigate the issues of high computational cost and insufficient real-time performance exhibited by traditional object detection models when deployed on edge devices[9].

In conclusion, research on waste detection has been conducted, but many challenges remain. For example, it's difficult to locate small objects, and the presence of other objects makes it even harder. As a result, the YOLO4SWD (YOLO with Smart Waste Detection) model, which is based on the YOLOv11 architecture, is suggested for finding trash. This paper makes the following contributions:

1. Innovatively design the CoordA module: This module allocates attention weights along the horizontal and vertical dimensions of waste objects, enabling accurate focusing on the local detailed features of small-sized waste and simultaneously enhancing the discriminability between low-contrast waste and background features.
2. Innovatively design the Basic Receptive Field Block (BasicRFB): This module is embedded into the feature fusion layer and adopts a multi-branch heterogeneous dilated convolution structure, which can effectively expand the model's receptive field. This design allows the model to not only accurately capture the local detailed features of waste objects but also fully cover their global contour information.
3. Propose the Feature Selection and Enhancement Attention Module (FSEAM) for detecting head optimization: An occlusion-aware attention mechanism is introduced into the YOLOv11 detection head to dynamically weight the features of occluded regions, assigning higher weights to valid unoccluded features and reducing the weight proportion of occluded region features, thereby significantly improving the waste classification accuracy in occluded scenarios.

The remainder of this paper is structured as follows: Sect. 2 reviews relevant research on waste detection techniques. Section 3 describes the architecture and technical details of the proposed model. Section 4 presents the experimental dataset, setup, evaluation metrics, and results. Section 5 discusses the experimental findings. Section 6 concludes the study and outlines future research directions.

Related Work

In contemporary waste management research, deep learning and data-driven intelligent solutions, waste detection, and classification have surfaced as pivotal challenges necessitating significant advancements. As sustainable development and smart city projects advance, researchers are focusing on using AI and sensor technologies to make waste sorting more accurate and efficient. Deep learning methods have made neural network-based waste recognition models very popular. The YOLOv8 model has an anchor-free detection structure, a custom loss function, and a feature restoration mechanism. It can correctly identify 97.7% of different types of waste, such as plastics, kitchen waste, batteries, and glass. It shows big improvements over YOLOv4, YOLOv5, YOLOv4, YOLOv5, YOLOv7, and older detection frameworks like SSD and Faster R-CNN. Because it can process data in real time, it is perfect for smart city applications[10]. The VGG16 model, which is based on a Convolutional Neural Network (CNN), has also been used in smart sorting systems for homes. It was able to sort items with 84.67% accuracy. This system uses a Raspberry Pi 400, ultrasonic sensors, and a camera to automatically tell the difference between paper, plastic, and glass. But its average accuracy shows that there is still room for improvement in network depth and feature extraction[11]. To address the constraints of single-modality visual recognition, certain studies have proposed multimodal learning strategies. One way to classify data across different modes is to combine image and audio data into a dual-stream neural network. This method uses Jetson Xavier NX and Raspberry Pi 4B to collect data in real time. It stays 96.24% accurate even when there is noise in the environment. This study shows that audio signals can help tell apart waste categories that look similar, making up for the flaws of traditional image-based models[12]. Another study suggested an IoT framework that links ultrasonic and humidity sensors to an Arduino Uno. This would allow for real-time monitoring and automatic sorting of wet and dry waste using LED indicators and servo motors. The system's dependence on particular primary waste categories constrains its applicability in intricate environments[13]. The Extent model uses a dual-stream Transformer architecture and the Data-efficient Image Transformer (DEiT) to accurately classify items like cell phones, laptops, and microwaves with an accuracy of 96%. Because it has fewer than 1 million parameters, it is light enough to work well in edge computing environments with limited resources[14]. Another study that used the GMC MobileNetV3 architecture used the CBAM attention module and Mish activation function. It was able to correctly classify kitchen waste and recyclables 96.55% of the time. This is a 3.6% improvement in performance over the old MobileNetV3, and it costs less to run[15]. To solve the problem of recognizing plastic waste, transfer learning methods combined with ResNeXt, DenseNet, and MobileNet_v2 models got 87.44% accuracy on the WaDaBa dataset, with ResNeXt being the best. This confirms the benefit of residual networks in classifying plastic waste[16].

With the rapid development of lightweight real-time detection algorithms, scholars have explored three mainstream technical routes to balance detection accuracy and computational overhead for waste detection tasks, including lightweight Transformer-based detectors, RT-DETR series lightweight models, and Mamba-based YOLO variants.

Lightweight Transformer detection models such as Swin-Tiny and DeiT-Tiny introduce global self-attention to capture long-distance contextual dependencies, which effectively alleviates feature confusion between waste objects and cluttered backgrounds. However, stacked multi-head attention modules bring excessive parameter volume and high GFLOPs. When deployed on low-computing platforms, their FPS drops sharply, making them hard to satisfy the real-time demands of on-site waste sorting. For garbage detection, most Transformer lightweight schemes only compress network depth and channel dimensions without targeted optimization for tiny, occluded waste targets, resulting in limited performance gain for small-size waste.

As an efficient real-time DETR architecture, RT-DETR abandons the non-maximum suppression (NMS) post-processing step and accelerates inference speed via uniform query matching. Light RT-DETR further shrinks the encoder and decoder structures to cut down model size. Nevertheless, the lightweight decoder still contains a large number of learnable queries, leading to higher Params and GFLOPs compared with tiny YOLO models. As shown in Table 4, Light RT-DETR reaches 4.5 M parameters and only 61 FPS, which lags behind tiny YOLO series in real-time performance under identical test conditions. It also lacks dedicated multi-scale branches for irregularly shaped waste, restricting its detection precision on fragmented tiny garbage.

In recent years, state space model Mamba has been combined with YOLO frameworks to construct Mamba-YOLO-tiny, replacing partial convolution layers with selective state space modules to model long-range features with lower computation cost than self-attention. Even so, Mamba-YOLO-tiny still has larger parameter volume and slower inference speed than YOLOv11n-based lightweight models. The linear state transition operation of Mamba easily loses fine-grained local features of micro waste, so it cannot achieve outstanding detection performance on small, low-contrast waste targets.

Compared with the above three types of lightweight detection frameworks, the YOLOv11n baseline adopted in this paper has a more concise decoupled structure, smaller native parameter overhead, and faster inference speed. On this basis, we design three task-oriented lightweight modules for tiny waste feature enhancement, multi-scale adaptation and occlusion suppression, achieving higher detection accuracy while maintaining competitive lightweight and real-time performance.Although YOLOv12n also achieves competitive accuracy, its internal structural coupling makes independent ablation of multi-scale and occlusion optimization modules difficult to implement, so YOLOv11n is selected as the baseline of this paper.

The ongoing development of deep learning and multimodal fusion technologies has greatly promoted the intelligent development of automatic waste recognition and sorting. Future research should concentrate on improving model adaptability, enhancing feature extraction for tiny and occluded waste, and constructing more efficient and robust waste detection frameworks. This work lays a solid foundation for intelligent waste detection based on the improved YOLOv11n model.

Proposed methods

Architecture of the proposed method

This research introduces a waste detection and classification algorithm called YOLO4SWD, derived from YOLOv11, designed to tackle the fundamental challenges of waste detection and classification in intricate scenarios. It specifically addresses critical challenges in practical applications, including substantial morphological variations of waste objects, uneven category distribution, and pronounced background interference (such as environmental clutter and lighting fluctuations), as well as feature misalignment and the loss of local mixed features due to inter-class occlusion, offering a detection and classification solution that reconciles efficiency and accuracy. We innovatively design a modified CoordA attention module. This module connects information about where things are in space with channel features. It gives different attention weights to the horizontal and vertical dimensions of waste targets. This allows for precise focus on the local details of small-sized waste while also making it easier to tell the difference between low-contrast waste and the background. This makes target recognition more robust and able to work in a wider range of situations. Then, the algorithm innovatively designs the BasicRFB module. Using a multi-branch heterogeneous dilated convolution structure, it makes the model's effective receptive field bigger. This design can capture both the global and local shapes of waste items while avoiding too much extra work for the computer. Lastly, to fix the problems of feature misalignment and loss caused by inter-class occlusion, an occlusion-aware attention mechanism is used to make the detection head (FSEAM) better. By changing the weight of the importance of features in occluded areas, it gives more importance to keeping valid features from unoccluded areas and less importance to information from occluded areas. This greatly improves the accuracy of waste classification in situations where it is blocked, making waste detection and classification more stable and efficient. The network architecture of YOLO4SWD is illustrated in Fig. 1.

CoordA module

Coordinate Attention (CoordA) is adopted and modified in this work to suppress background interference in cluttered waste scenes and strengthen fine-grained edge feature extraction for tiny waste targets. Compared with the original Coordinate Attention module proposed in[17], three targeted optimizations tailored to waste detection characteristics are implemented as follows:

The two separate 1D global pooling branches in the original structure are replaced by shared pooling layers to eliminate redundant feature calculation and reduce extra computational overhead.

A lightweight feature filtering branch is inserted before coordinate encoding to restrain invalid background texture responses and alleviate feature confusion between waste objects and complicated backgrounds.

The channel attention activation function is changed from Sigmoid to Hard-Swish, which preserves faint edge features of fragmented tiny waste without adding extra parameters.

Ablation experiment results further validate the effectiveness of the improved CoordA. After embedding this module alone into YOLOv11n, the total model parameters only increase slightly from 2.58 M to 2.62 M while GFLOPs remains 6.3, with FPS marginally dropping from 84 to 81. For detection performance, precision rises from 0.813 to 0.814, recall improves from 0.748 to 0.751, and mAP@0.5 gains a 1.3% increment to reach 0.801. These results prove that the optimized CoordA brings steady performance improvement with almost no loss of lightweight property, making it applicable to real-time waste detection tasks. The structural flowchart of this module is displayed in Fig. 2.

As illustrated in Fig. 2, the improved CoordA first introduces a lightweight feature filtering branch to suppress background texture interference before coordinate encoding. Subsequently, shared coordinate pooling extracts

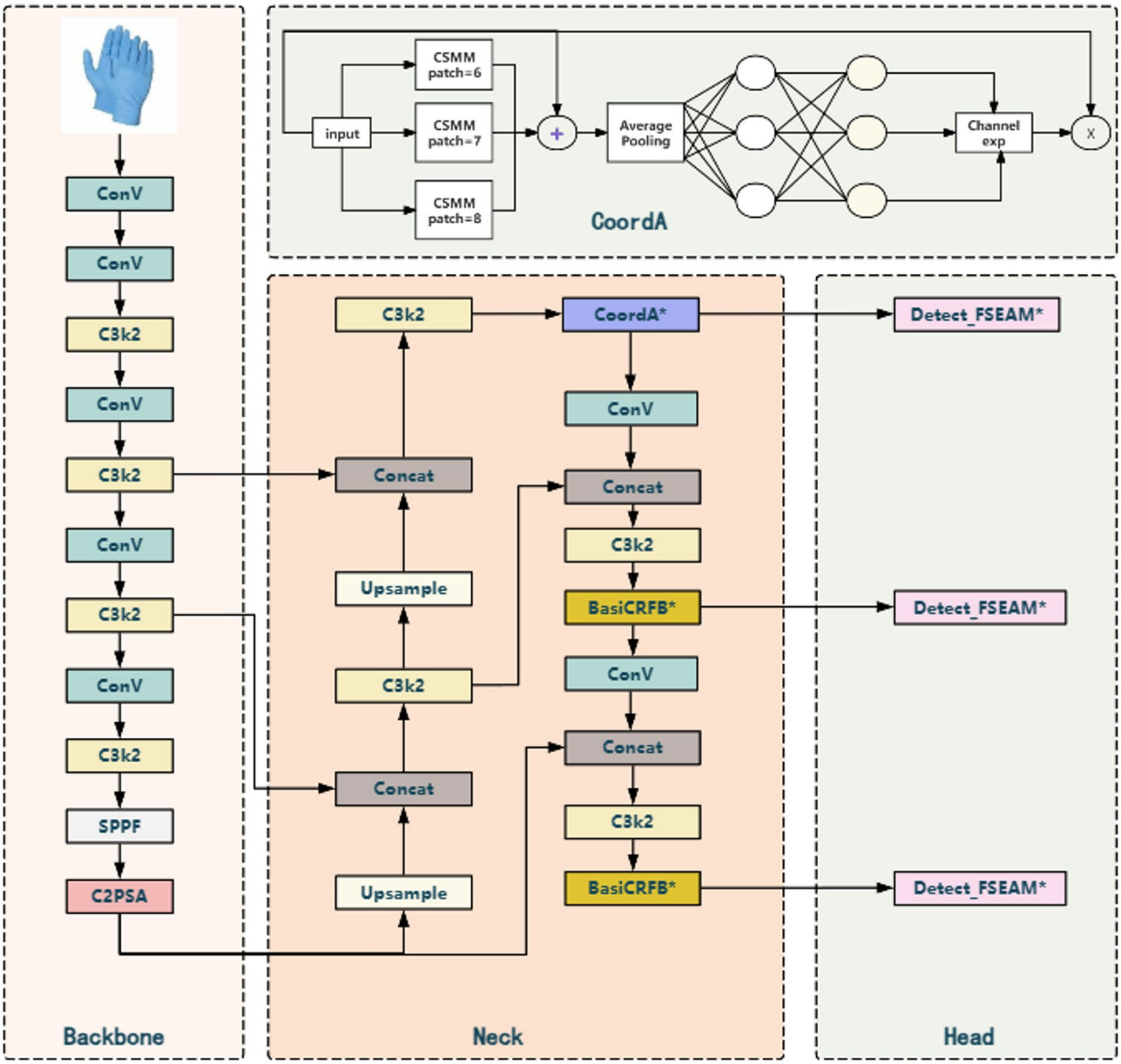


Fig. 1. The structure of the YOLO4SWD network, with the modules identified by an asterisk (*) representing the key improvements of this study.

long-range contextual information along horizontal and vertical directions. The pooled features are fused and compressed through a shared convolution layer followed by Hard-Swish activation. Finally, two directional attention maps are generated and applied to the original feature map for adaptive feature enhancement. The corresponding mathematical formulations are presented in Eqs. (1)–(7).

Coordinate information embedding

Global average pooling in conventional channel attention squeezes global spatial information into channel descriptors, which inevitably loses precise positional cues. In this work, the single global pooling operation is decomposed into two independent one-dimensional feature encoding branches, enabling the attention module to capture long-range spatial dependencies with accurate positional information. The relevant formulas are defined as follows:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_c(i, j) \tag{1}$$

z_c: represents the result of applying Global Average Pooling to the feature map of the c-th channel. It is a scalar value that captures the global statistical information of the features in that channel. $x_c(i, j)$: denotes the feature

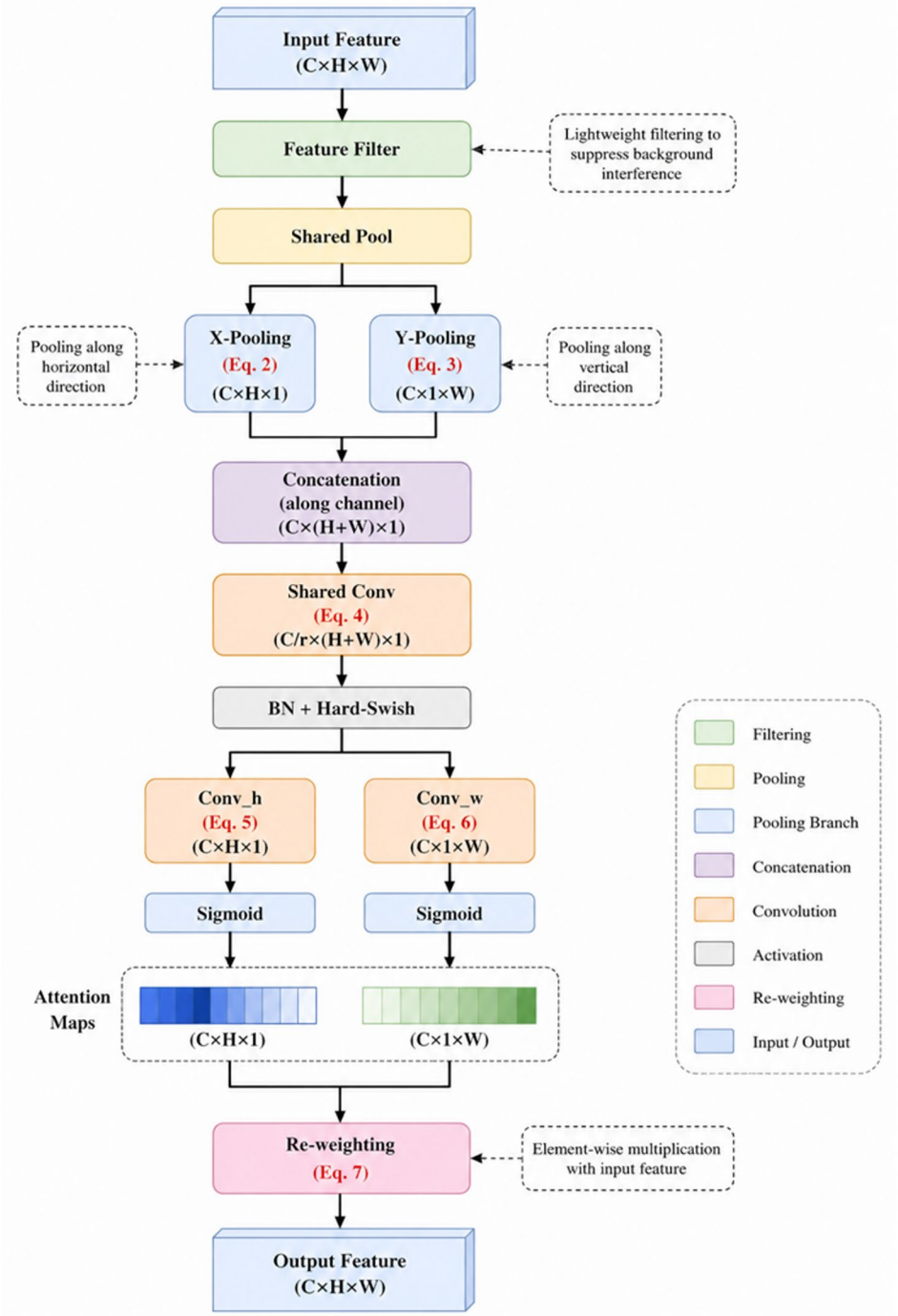


Fig. 2. Visualized workflow of the proposed improved CoordA module.

value at the spatial location (i, j) in the feature map of the c-th channel.H: The height of the feature map (number of vertical pixels).W: The width of the feature map (number of horizontal pixels). $\sum_{i=1}^{H}\sum_{j=1}^{W} x_c(i,j)$: The summation is done over all the spatial positions of the feature map for the c-th channel. This is done to combine the channel's global features. $\frac{1}{H\times W}$: The summation result is divided by the total number of pixels in the feature map, performing an averaging operation to obtain the mean value of the features in that channel, which constitutes the final output of the Global Average Pooling operation.

The output of the c-th channel with height h can be represented as:

$$z_c^h(h) = \frac{1}{W}\sum_{0\le i<W} x_c(h,i) \tag{2}$$

The output of the c-th channel with width w can be written as:

$$z_c^w(w) = \frac{1}{H} \sum_{0 \le j < H} x_c(j, w) \tag{3}$$

Coordinate Attention Generation

Before coordinate encoding, a lightweight feature filtering branch is introduced to suppress invalid background texture responses and reduce feature confusion between waste targets and complex backgrounds.

The core procedure extracts position-aware features from horizontal and vertical dimensions simultaneously. The model first performs pooling along row and column axes, then concatenates and transforms the pooled features. After channel-wise splitting and activation transformation, directional attention weights for horizontal and vertical directions are generated to reweight the original feature map. This mechanism guides the network to focus on target-related regions while suppressing irrelevant background features.

$$f = \delta\ (F_1([z^h, z^w]\,)) \tag{4}$$

$\mathbf{z^h}$ The result obtained by performing global average pooling on the feature map along the horizontal dimension (height) focuses on the statistical information of each row.

$\mathbf{z^w}$ The result obtained by performing global average pooling along the vertical dimension (width) focuses on the statistical information of each column.

$\mathbf{z^h}$, $\mathbf{z^w}$ The pooling results from the horizontal and vertical directions are concatenated along the channeldimension to fuse spatial features.

$\mathbf{F_1}$ A transformation function performs dimensionalityreduction and information integration on the concatenated features, learning spatial correlations.

$\boldsymbol{\delta}$ An activation function introduces nonlinearity to enhance expressive power.

$\mathbf{f}$ The resulting intermediate feature vector, fusing horizontal and vertical spatial information, serves as the foundation for subsequent attention weight generation.

$$g^h = \sigma\ (F_h(f^h)) \tag{5}$$

$$g^w = \sigma\ (F_w(f^w)) \tag{6}$$

$\mathbf{f^h}$, $\mathbf{f^w}$ Split the intermediate features along the channel dimension to obtain sub-feature vectors related to the horizontal and vertical directions, respectively.

g^h, g^wPerform targeted dimensionality transformation and information refinement on the horizontal and vertical sub-feature vectors.

The resulting horizontal and vertical attention weights are used to apply weighting to the horizontal and vertical dimensions of the original feature map.

Finally, the output of the CoordA module can be written as:

$$y_c(i, j) = x_c(i, j) \times g_c^h(i) \times g_c^w(j) \tag{7}$$

BasicRFB module

To address the large scale span of domestic waste targets ranging from tiny fragmented debris to large-volume waste containers, the original RFB module is simplified and lightweighted in this paper, named BasicRFB. Compared with the original RFB structure proposed in[18], two lightweight optimizations are implemented for waste detection scenarios:

Remove the redundant multi-branch large-dilation convolution branch that brings massive parameters; retain only three groups of small and medium receptive field branches to balance multi-scale feature extraction and computational cost.

Replace all standard 3×3 convolutions inside branches with depthwise separable convolution, which greatly cuts down parameter volume and GFLOPs without weakening receptive field feature aggregation ability.

This lightweight BasicRFB can simultaneously extract feature information of small, medium and large waste targets, effectively solving the problem of insufficient multi-scale perception in the baseline YOLOv11n. Ablation experiments verify its performance gain on recall metric. After adding BasicRFB to YOLOv11n alone, model parameters slightly rise from 2.58 M to 2.62 M with GFLOPs maintained at 6.3, and FPS only drops from 84 to 81. In detection metrics, recall is improved from 0.748 to 0.751, and mAP@0.5 increases by 1.3% to 0.801. These results demonstrate that BasicRFB achieves obvious recall improvement for multi-size waste targets at the cost of negligible lightweight loss. The structural diagram of BasicRFB is shown in Fig. 3.

BasicRFB extracts multi-scale features through three parallel branches with distinct convolution configurations, each responsible for capturing features at different receptive field ranges. Different from the original RFB, redundant branches with ultra-large dilation rates are removed, and depthwise separable convolution replaces standard convolution within each branch to achieve lightweight optimization.

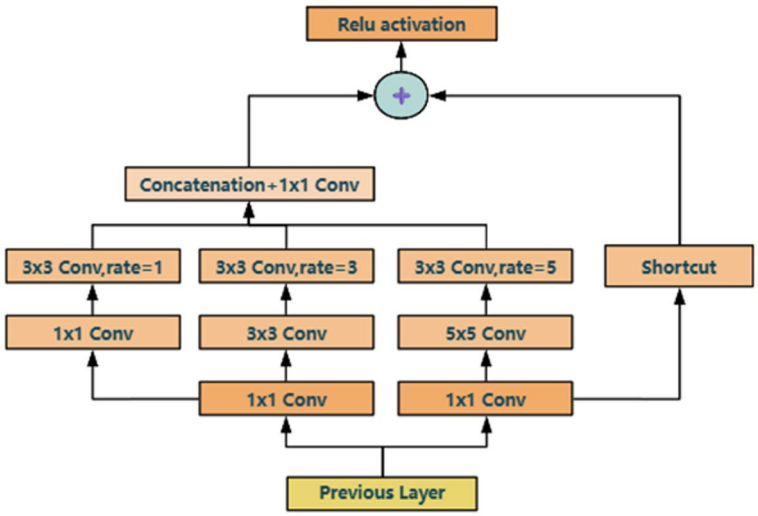


Fig. 3. Algorithm flowchart of BasicRFB.

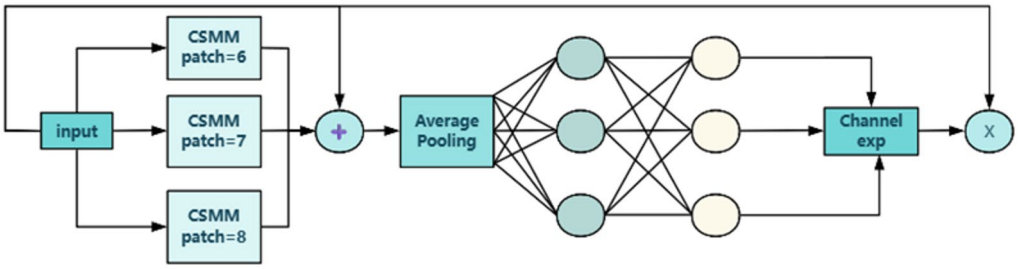


Fig. 4. FSEAM Module architecture diagram.

The first branch (left) firstly reduces channel dimensions via 1×1 convolution, followed by undilated 3×3 convolution (dilation rate = 1) to capture fine-grained local details of tiny waste targets.

Branch 2 (middle) adopts successive 1×1 and 3×3 convolutions to extract fundamental features, then employs dilated 3×3 convolution (dilation rate = 3) to expand receptive fields. This branch balances local texture features and medium-range contextual information.

For Branch 3 (rightmost), 1×1 and 5×5 convolutions are used to extract high-level semantic features, and a dilated 3×3 convolution with dilation rate = 5 is applied to acquire global contextual information of large-size waste.

Feature Fusion and Enhancement (Top Part): Multiscale outputs from three branches are concatenated along the channel dimension, integrating fine details, medium-range textures and global context to form a comprehensive multi-scale feature representation. After concatenation, a 1×1 convolution layer is utilized for channel dimension compression. A residual shortcut branch is introduced to directly transmit original input features to the output end. The fused multi-scale features are element-wise added with the raw input features, and an activation function is applied to introduce nonlinear representation capability.

FSEAM detection head

This paper designs the FSEAM detection head to mitigate missed detection caused by occluded waste. Drawing on optimization ideas from face detection tasks[19], feature attention mechanism is embedded into the detection head to boost overall detection performance, with two core objectives: capturing multi-scale waste features and suppressing irrelevant background interference. Targeted structural design enables the module to reconstruct incomplete target features and filter background noise under occlusion scenarios. The overall architecture of the FSEAM module is illustrated in Fig. 4.

The front-end of the module adopts depthwise separable convolution[20] combined with residual connection structure[21]. Feature extraction is realized through two-stage depthwise convolution[22] and pointwise convolution[23]. To strengthen global feature correlation and explicitly distinguish feature differences between occluded and complete waste targets, two fully connected layers are arranged after depthwise separable convolution. The first fully connected layer flattens the 2D feature map into a 1D feature vector and aggregates global statistical information along the channel dimension. The second fully connected layer introduces nonlinear transformation

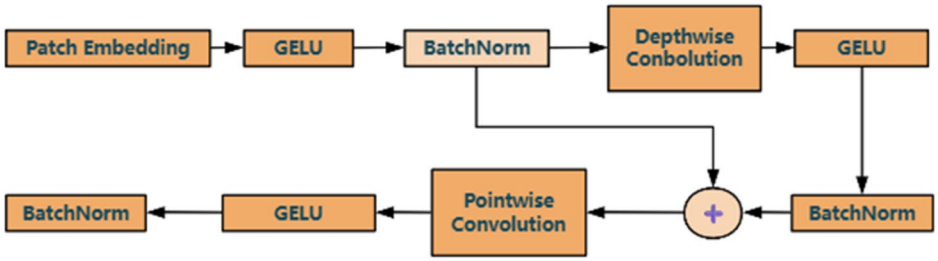


Fig. 5. Structure diagram of CSMM.

Environmental Parameters	Value
Operating System	Ubuntu 22.04
CPU	Xeon(R) Platinum 8470Q
GPU	RTX4090 (24 GB)
Programming Language	Python3.11
Deep Learning Framework	Pytorch2.01
RAM	16 GB

Table 1. Environment configuration table.

Image Size	Batch	Cycle	Optimizer	Learning Rate
640	32	200	SGD	0.01

Table 2. Experiment Configuration Table.

to learn weight correlation between occluded and visible feature regions, so that the model can infer missing content of blocked areas based on intact local features and avoid loss of blended texture information.

Exponential normalization is applied to the output of fully connected layers to resolve feature alignment errors induced by inter-object occlusion, such as fuzzy target boundaries and positional offset between occluders and waste objects. Finally, the original feature map is multiplied by the normalized attention weights for feature enhancement: high-weight regions corresponding to effective waste targets are highlighted, while invalid background and heavily occluded low-response regions are suppressed.

Ablation experiments verify the lightweight and performance advantages of FSEAM. When the module is solely introduced into YOLOv11n, model parameters slightly rise from 2.58 M to 2.62 M, GFLOPs remains 6.3 without additional computational overhead, and FPS only slightly decreases from 84 to 81. For detection accuracy, recall is significantly improved from 0.748 to 0.781, and mAP@0.5 increases by 2.3% to 0.811, proving that FSEAM effectively reduces missed detection of occluded waste without extra computation cost.

The Channel and Spatial Mixing Module (CSMM) is constructed to strengthen feature representation capacity. It conducts joint feature optimization on channel and spatial dimensions simultaneously to lift the model's discrimination ability for waste targets. The structural layout of CSMM is displayed in Fig. 5:

Experiment

Experimental environment

The detailed configuration information for the environment used in this paper is provided in Table 1, while the experimental environment configuration is detailed in Table 2.

Dataset

This paper employs data from a publicly accessible online dataset[24] containing 7980 images.There are 16 types of datasets, such as batteries, Organic waste, Cardboard, Glass, Glass Bottles, Gloves, High-density Polyethylene Plastic, Masks, Pharmaceuticals, Newspapers, PET bottles, Paper, Single-use Plastic, Single-layer Plastic, Extruded tubes, and Syringes. The dataset is divided into three parts: a training set (75%), a validation set (15%), and a test set (15%).Sample images from these datasets are shown in Fig. 6.

Evaluation indicators

This paper uses precision, recall, F_1 score, and mean average precision (mAP) as key metrics to measure how well the proposed model works. One of the most important ways to measure how well a model works is by looking at its precision. It tells you how many of the model's positive predictions were right. This shows how accurate the model's positive predictions are, and it is calculated like this:

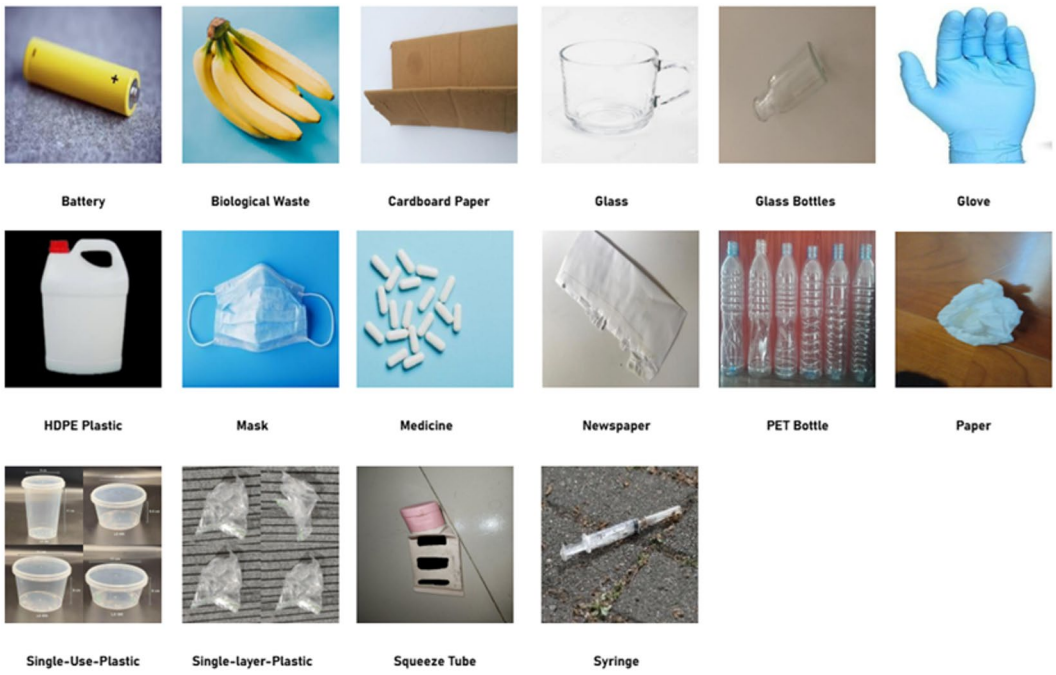


Fig. 6. Dataset.

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{8}$$

Recall tells you how many of the positive samples the model correctly identifies.It measures how well the model can find positive samples by looking at the percentage of true positives that were correctly recognized.This is what it means:

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{9}$$

False Negative (FN) : The number of positive samples that were incorrectly predicted to be negative.

The F_1 score is the harmonic mean of precision and recall. It is a full measure of how well a model balances its ability to accurately predict positive classes and cover all positive classes.It is especially useful when datasets are not balanced.The formula for its calculation is:

$$\text{F}_1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{10}$$

To get Average Precision (AP), you take the precision at each position i and multiply it by the increase in recall at that position.Then you add up all of these products.The area under the precision-recall curve, which is based on the intersection-over-union score between predicted and actual boxes, is how it is calculated.It is defined as:

$$\text{AP} = \sum {}^{\text{n}}_{\text{i}=1} \text{P}\,(\text{i}) \cdot \Delta\,\text{R}\,(\text{i}) \tag{11}$$

P(i) is the accuracy rate at the i-th threshold, and ΔR(i) is the change in the recall rate at that threshold.

The mean average precision (mAP) is the average value of the average precision (AP) across all categories. Specifically, mAP@0.5 refers to the AP calculated when the Intersection over Union (IoU) threshold is set to 0.50, while mAP@0.75 refers to the mAP calculated when the IoU threshold is set to 0.75. In contrast, mAP@0.5:0.95 is obtained by averaging the AP values across multiple IoU thresholds ranging from 0.50 to 0.95. The formula is as follows:

$$\text{mAP} = \frac{1}{\text{n}} \sum {}^{\text{n}}_{\text{c}=1} \text{AP}_{\text{c}} \tag{12}$$

In this case, n stands for the total number of classes, and APc stands for the average precision for class c.

Ablation experiment

This paper conducts ablation experiments to quantitatively assess the contribution of each proposed module and to eliminate mutual interference among modules, establishing evaluation metrics based on YOLOv11n and inter-

module evaluation metrics. Evaluation metrics in Table 3 are P, R, F_1, mAP@0.5, mAP@0.75, mAP@0.5:0.95, Params(M), GFLOPS, and FPS.

Compared with the baseline YOLOv11n ($P = 81.3\%$, $R = 74.8\%$, $F_1 = 77.4\%$, Params = 2.58 M, GFLOPs = 6.3, FPS = 84), the standalone integration of the CoordA module leads to slight improvements in P, R, and F_1 (reaching 81.4%, 75.1%, and 77.5%, respectively), with mAP@0.5 and mAP@0.75 increasing by 1.3 and 1.8% points (to 80.1% and 68.5%, respectively). This lightweight module only adds 0.04 M parameters without extra computational overhead (GFLOPs remains 6.3), and FPS slightly drops to 81. It can enhance the distinction between edge textures and the background, reduce feature confusion, and lay the groundwork for subsequent optimization.

When the BasicRFB module is added independently, R rises by 1.3% points from 74.8% to 76.1%, F_1 reaches 78.1%, and mAP@0.5 increases by 1.7% points to 80.5%. This multi-scale module brings mild parameter growth to 2.65 M and raises GFLOPs to 6.8, resulting in FPS decreasing to 79. It is adaptable to the diverse morphologies and large size ranges of waste, improving recall while maintaining stable precision.

The FSEAM module, when acting alone, delivers the most significant recall gain among single-module configurations: R increases by 3.5% points from 74.8% to 78.3%, F_1 rises by 1.2% points to 78.6%, and mAP@0.75 hits 69.2%. Notably, FSEAM optimizes calculation efficiency, cutting GFLOPs down to 5.9 with only a tiny parameter increment (2.63 M), and FPS only falls to 80. It can alleviate feature loss caused by inter-class occlusion of waste and enhance the model's detection robustness in complex stacked scenarios.

The synergistic effects of module combinations and their lightweight overhead are analyzed as follows:

After combining CoordA with BasicRFB, the precision (P) reaches 83.3% (2% points higher than the baseline), with both F_1 (78.9%) and mAP@0.5 (81.1%) exceeding those of single-module configurations. The combined model has Params = 2.69 M, GFLOPs reduced to 5.9 and FPS = 77. The two modules avoid redundancy through the "feature filtering—multi-scale expansion" pipeline and improve input feature reliability.

The combination of CoordA and FSEAM achieves a precision (P) of 84.8% (3.5% points higher than the baseline) while maintaining a high mAP@0.5 (81.2%). Its Params and GFLOPs stay at 2.67 M and 6.3, with FPS dropping to 78, which reflects the strong complementarity between foreground feature focusing and occlusion suppression.

The combination of BasicRFB and FSEAM yields the highest precision ($P = 84.9\%$) across all ablation groups, with mAP@0.75 (68.6%) far exceeding the baseline. The dual-module setup slightly increases parameters to 2.71 M while keeping GFLOPs at 6.3 and FPS = 76, showing superior performance on waste objects with moderate-to-low occlusion and variable sizes.

When all three modules are integrated into the complete YOLO4SWD model, the F_1-score rises to 79.9%, mAP@0.5 reaches 81.6%, mAP@0.75 peaks at 69.9%, and mAP@0.5:0.95 improves to 62.5%. In terms of complexity, the full model only adds 0.16 M parameters relative to the baseline (Params = 2.74 M) with unchanged GFLOPs of 6.3, and retains a real-time FPS of 74. The three-stage collaborative design (background filtering via CoordA, multi-scale feature enhancement via BasicRFB, occlusion-aware feature optimization via FSEAM) systematically addresses core pain points of waste detection, including cluttered backgrounds, irregular object shapes, and frequent target occlusion. Even with negligible extra computation cost, the proposed pipeline greatly boosts multi-scale detection accuracy for 16 waste categories, which fully verifies the effectiveness and lightweight advantage of our improved strategy.

Comparative experiment

To further verify the practicability of our waste detection model, we compare it with mainstream advanced detectors on the unified test set, as shown in Table 4. We use precision, recall, F_1-score and multi-threshold mAP to measure detection accuracy. We also add Params(M), GFLOPs and FPS to evaluate the model's lightweight property and inference speed.

YOLOv5n and YOLOv9n exhibit the weakest overall detection performance. Their precision only reaches 0.575 and 0.612 respectively, recall stays below 0.6, F_1-scores are less than 0.6, and they obtain the lowest multi-threshold mAP, with mAP@0.5:0.95 merely 0.402 and 0.437. Such poor metrics indicate they cannot satisfy the practical demands of waste detection. Two-stage detectors ResNet50-Faster[25] and SSD[26] achieve moderate accuracy but suffer from extremely large model overhead: their Params reach 36.2 M and 24.7 M, GFLOPs hit 187.5 and 126.3, and FPS drops to only 12 and 18, which makes them unsuitable for lightweight real-time scenarios.

YOLOv11n	CoordA	BasicRFB	FSEAM	*P*	*R*	F_1	mAP@0.5	mAP@0.75	mAP@0.5:0.95	Params (M)	GFLOPs	FPS
✓				0.813	0.748	0.774	0.788	0.662	0.599	2.58	6.3	84
✓	✓			0.814	0.751	0.775	0.801	0.685	0.617	2.62	6.3	81
✓		✓		0.816	0.761	0.781	0.805	0.675	0.618	2.65	6.8	79
✓			✓	0.801	0.783	0.786	0.805	0.692	0.614	2.63	5.9	80
✓	✓	✓		0.833	0.759	0.789	0.811	0.692	0.625	2.69	5.9	77
✓	✓		✓	0.848	0.749	0.785	0.812	0.673	0.613	2.67	6.3	78
✓		✓	✓	0.849	0.717	0.761	0.803	0.686	0.612	2.71	6.3	76
✓	✓	✓	✓	0.829	0.781	0.799	0.816	0.699	0.625	2.74	6.3	74

Table 3. Ablation results of each improved module based on YOLOv11n.

Model	*P*	*R*	F_1	mAP@0.5	mAP@0.75	mAP@0.5:0.95	Params (M)	GFLOPs	FPS
ResNet50-Faster	0.717	0.656	0.673	0.648	0.522	0.482	36.2	187.5	12
SSD	0.757	0.636	0.642	0.661	0.542	0.502	24.7	126.3	18
YOLOv5n	0.575	0.517	0.542	0.568	0.436	0.402	1.9	4.2	92
YOLOv8n	0.771	0.715	0.735	0.763	0.652	0.577	3.4	7.8	76
YOLOv9n	0.612	0.571	0.571	0.572	0.471	0.437	2.8	6.5	81
YOLOv10n	0.714	0.665	0.682	0.706	0.612	0.538	2.1	5.1	88
YOLOv11n	0.813	0.748	0.774	0.788	0.662	0.599	2.58	6.3	84
YOLOv12n	0.801	0.736	0.761	0.772	0.653	0.581	2.66	6.4	80
Mamba-YOLO-tiny	0.805	0.74	0.765	0.778	0.657	0.586	3.1	7.2	72
Light RT-DETR	0.808	0.742	0.768	0.781	0.66	0.59	4.5	9.6	61
Ours	0.829	0.781	0.799	0.816	0.699	0.625	2.74	6.3	74

Table 4. Performance evaluation of different models for waste detection.

YOLOv8n, YOLOv10n[27], YOLOv12n[28], Mamba-YOLO-tiny and Light RT-DETR deliver improved detection accuracy, with all F_1-scores exceeding 0.7 and mAP@0.5 above 0.7. Among them, YOLOv8n gains a mAP@0.5:0.95 of 0.577, yet its 3.4 M parameters and 7.8 GFLOPs bring extra computational cost. Mamba-YOLO-tiny and Light RT-DETR possess competitive precision but carry heavier model weights and slower inference speeds, with FPS only 72 and 61 separately.

Our proposed model outperforms all contrastive models on every accuracy indicator while retaining lightweight characteristics. It achieves precision of 0.829, 0.016 higher than baseline YOLOv11n to lower false detection rates; recall reaches 0.781, an improvement of 0.033 over YOLOv11n that effectively reduces missed waste targets; the F_1-score rises to 0.799, realizing a better trade-off between precision and recall. For multi-IoU mAP metrics, our model obtains mAP@0.5 = 0.816, mAP@0.75 = 0.699 and mAP@0.5:0.95 = 0.625, 0.028, 0.037 and 0.026 higher than YOLOv11n respectively, proving strong adaptability to diverse detection difficulty thresholds.

In terms of lightweight performance, our model only increases parameters to 2.74 M with unchanged GFLOPs of 6.3 compared with YOLOv11n (2.58 M, 6.3 GFLOPs), and maintains a real-time FPS of 74. To sum up, the proposed model achieves outstanding detection precision, comprehensive target capture ability and favorable lightweight performance simultaneously, making it more competent for real-world waste detection tasks.

Visualization

To intuitively reveal the performance improvements brought by our optimized modules under complex real-world waste scenarios, two categories of visual comparative analyses are conducted in this section. We present three groups of detection comparison charts corresponding to three challenging waste cases: micro fragmented waste, heavily occluded waste, and targets amid cluttered outdoor backgrounds. By comparing prediction results between vanilla YOLOv11n and the fully optimized YOLO4SWD, we can visually observe the obvious suppression of missed detections and false positives introduced by the three embedded modules. As illustrated in Fig. 7, these contrast results verify that the integrated optimization strategy can effectively reduce missed targets, elevate prediction confidence, and reconstruct missing feature information for occluded and micro-sized waste in complicated field environments.

Cross-dataset generalization experiment

To verify the generalization ability of the improved model on external datasets beyond the self-built domestic waste dataset, two public standard waste datasets, WaDaBa and TrashNet, are adopted for cross-dataset testing in this section.For fair comparison, all experimental hyperparameters, input resolution and test hardware remain consistent with previous experiments. We take the original YOLOv11n as the baseline model and compare it with the final improved model equipped with CoordA, BasicRFB and FSEAM modules. mAP@0.5 are recorded as quantitative evaluation metrics.

Table 5 lists the detection results on WaDaBa[29] and TrashNet[30]. On both public datasets, the improved model achieves obvious mAP promotion compared with the baseline YOLOv11n. It demonstrates that the three optimized modules proposed in this paper do not only bring performance gains for the self-collected dataset, but also possess strong generalization capability for various types of waste images collected under different shooting environments, illumination conditions and background complexities. The multi-scale feature extraction, tiny object enhancement and occlusion feature reconstruction strategies can be effectively migrated to different waste detection scenarios.

Occlusion group contrast experiment

To quantitatively evaluate the anti-occlusion performance of FSEAM proposed in Sect. 3.4, the test set is divided into four groups according to the target occlusion ratio: 0%, 20%, 40% and 60%. Two comparison models are tested separately: the baseline YOLOv11n without FSEAM and the single-branch model equipped only with FSEAM. mAP@0.5 is adopted as the core evaluation metric to quantify the performance improvement brought by FSEAM under different occlusion degrees.

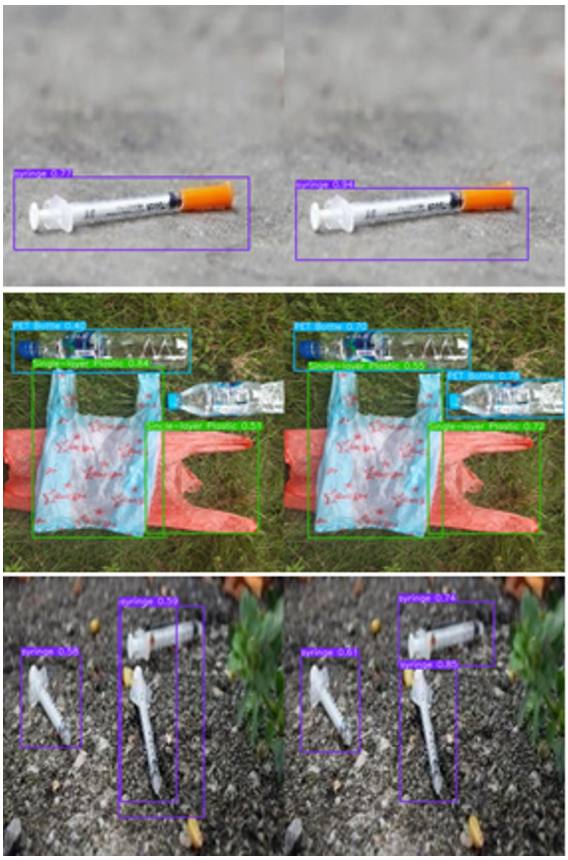


Fig. 7. Detection rendering comparison under three typical complex waste detection scenes. Left columns: prediction outputs of baseline YOLOv11n; Right columns: prediction outputs of the proposed YOLO4SWD model. Row 1: Single tiny syringe medical waste on smooth concrete ground; Row 2: Overlapped PET bottles and plastic bags placed on the lawn; Row 3: Multiple tiny syringes mixed with weeds and gravel in a cluttered outdoor background.

Model	WaDaBa mAP@0.5	TrashNet mAP@0.5
YOLOv11n	0.762	0.785
Ours	0.795	0.817

Table 5. mAP@0.5 Comparison of baseline YOLOv11n and the Proposed Model on WaDaBa and TrashNet Datasets.

Table 6 lists the mAP indicators of the two models under four occlusion levels. With the increase of occlusion ratio, the mAP of the baseline model drops sharply due to serious loss of target effective features. By contrast, the model embedded with FSEAM maintains relatively stable detection accuracy. Under heavy occlusion of 60%, the mAP of the model with FSEAM is 3.9% higher than that of the baseline.

The above results verify that FSEAM can effectively reconstruct incomplete feature information of occluded waste and reduce missed detection, which is consistent with the module design motivation described in Sect. 3.4.

Multi-scale target test

To verify the multi-scale feature extraction capability of the lightweight BasicRFB module proposed in Sect. 3.3, all waste samples in the test set are divided into three categories: small, medium and large targets according to the pixel area of bounding boxes. We compare the baseline YOLOv11n and the model only equipped with BasicRFB, and record mAP@0.5 for each scale category to quantify the detection improvement across different target sizes.

Table 7 lists the mAP@0.5 results of the two models on three scale groups. For small-sized waste fragments, the baseline suffers from insufficient receptive field perception and low detection accuracy. After adding BasicRFB, mAP@0.5 increases obviously. For medium and large waste objects, the model with BasicRFB also achieves steady mAP gains.

Occlusion Ratio	Model	mAP@0.5
0%	YOLOv11n	0.831
0%	FSEAM	0.837
20%	YOLOv11n	0.794
20%	FSEAM	0.82
40%	YOLOv11n	0.752
40%	FSEAM	0.79
60%	YOLOv11n	0.686
60%	FSEAM	0.725

Table 6. mAP@0.5 comparison under different target occlusion ratios.

Target Scale	Model	mAP@0.5
Small	YOLOv11n	0.724
Small	BasicRFB	0.768
Medium	YOLOv11n	0.835
Medium	BasicRFB	0.861
Large	YOLOv11n	0.872
Large	BasicRFB	0.89

Table 7. mAP@0.5 comparison for small/medium/large multi-scale waste targets.

The above results prove that the multi-branch receptive field design of BasicRFB effectively adapts to domestic waste targets with a wide size span, consistent with the lightweight multi-scale optimization goal described in Sect. 3.3.

Module feature response visualization

To further analyze the internal feature optimization mechanism of the three proposed modules, Class Activation Mapping (CAM) heatmaps are adopted for ablation visualization analysis in this subsection. We conduct ablation comparisons among four model variants: vanilla YOLOv11n baseline, baseline equipped solely with CoordA, baseline equipped solely with BasicRFB, and baseline equipped solely with FSEAM.

Figure 8 presents CAM heatmap comparisons corresponding to three typical waste scenes. For the vanilla YOLOv11n baseline, the feature activation areas are scattered, incomplete, and severely disturbed by irrelevant background textures. After integrating each optimized module separately, distinct improvement characteristics can be observed: CoordA concentrates high-intensity activation on the fine edges of tiny fragmented waste; BasicRFB expands the effective multi-scale receptive field to fully cover waste targets of different sizes; FSEAM reconstructs missing feature responses for mutually occluded waste objects.

The above visual results collectively verify the independent design objectives of CoordA, BasicRFB and FSEAM proposed in Sect. 3, which demonstrates that each module achieves its expected feature optimization function for difficult waste detection scenarios.

Discussion

This work takes YOLOv11n as the baseline model and carries out targeted structural optimization for domestic waste detection, and comprehensive quantitative evaluation is conducted on the self-built waste dataset. Compared with the original YOLOv11n, the optimized model achieves consistent gains across multiple core metrics: recall rises from 74.8% to 78.1% (+3.3%), F_1-score increases from 77.4% to 79.9% (+2.5%), mAP@0.5 improves from 78.8% to 81.6% (+2.8%), mAP@0.75 grows from 66.2% to 69.9% (+3.7%), and mAP@0.5:0.95 increases from 59.9% to 62.5% (+2.6%). When compared with the lightweight YOLOv12n, the optimized model gains precision (82.9% vs. 80.1%, +2.8%), recall (78.1% vs. 73.6%, +4.5%) and mAP@0.5:0.95 (62.5% vs. 58.1%, +4.4%). The above results reveal that the proposed optimization strategy delivers steady performance promotion, and the improvement becomes more prominent under strict IoU matching thresholds.

Three customized modules are embedded to address typical challenges in waste detection scenes. The CoordA module is designed to suppress false positives and missed detection under variable illumination and cluttered backgrounds, strengthening the model's adaptability to real sanitation scenarios. To compensate for insufficient receptive field perception of tiny targets in the original backbone-neck structure, the lightweight BasicRFB module is introduced into the neck feature fusion branch, supplying complete multi-scale feature representations for subsequent detection heads. Furthermore, the FSEAM module alleviates feature dislocation and incomplete local information loss induced by inter-object occlusion, lifting classification accuracy for blocked waste instances. Jointly, the three modules enhance the model's capability for complex backgrounds, multi-size targets and occluded samples.

Nevertheless, several limitations remain in the optimized model. Transparent objects such as glass bottles produce uneven reflective features, which are easily confused with transparent plastic bag backgrounds. In

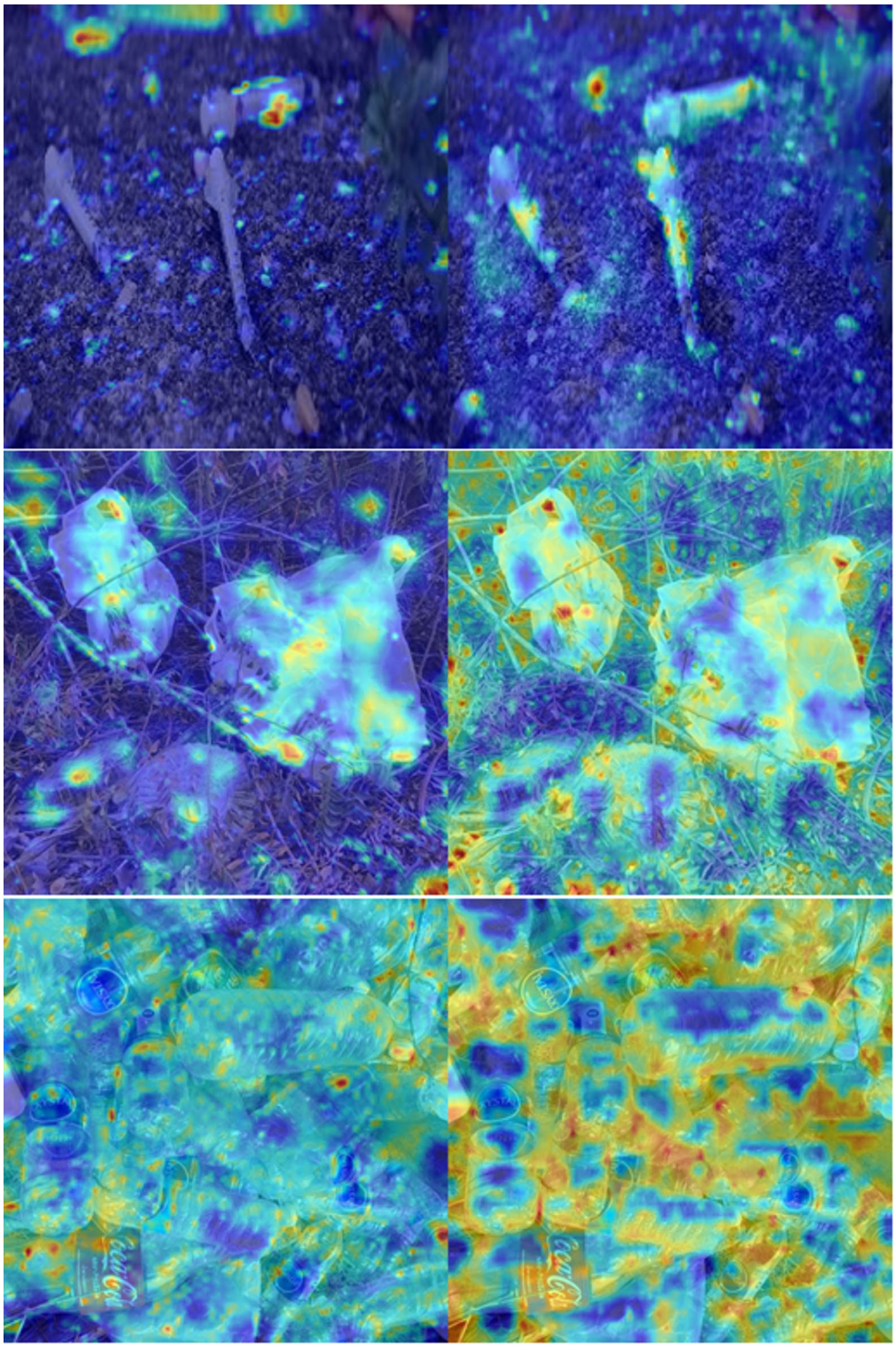


Fig. 8. CAM heatmap ablation comparison of three optimized modules. Left column: feature activation heatmap of vanilla YOLOv11n baseline; Right column: heatmap of baseline embedded with a single improved module. Row 1: Multiple tiny syringes mixed with weeds for CoordA verification; Row 2: Large plastic bags in wild weeds for BasicRFB verification; Row 3: Densely stacked occluded plastic bottles for FSEAM verification. The three subgraphs intuitively reveal the independent optimization function of each module on micro targets, large-size targets and occluded targets respectively.

addition, the bounding box localization accuracy for flexible targets like disposable masks still needs further improvement. These phenomena indicate that the feature extraction and generalization capacity of the model still have room for further optimization.

Conclusions

This paper proposes an optimized waste detection model named YOLO4SWD based on YOLOv11n, aiming to tackle three prevalent drawbacks of lightweight detectors in domestic waste identification: high missed detection rate of tiny fragments, weak robustness under complex backgrounds, and degraded classification accuracy for occluded targets. The optimized model can provide a feasible technical solution for intelligent waste sorting and automatic classification equipment.

Three dedicated modules are integrated into the YOLO4SWD pipeline. The CoordA module enhances discriminative local feature extraction for small low-contrast waste fragments to reduce background interference. The lightweight BasicRFB module is embedded in the neck fusion layer, which adopts multi-branch dilated convolution to expand the effective receptive field while constraining computational overhead, adapting to waste targets with large size variation ranges. The FSEAM module is deployed at the detection head to mitigate feature loss caused by mutual occlusion between different waste categories and stabilize classification performance under blocked scenes.

Quantitative results on the test set show that the optimized YOLO4SWD achieves mAP@0.5 of 81.6%, mAP@0.75 of 69.9%, and mAP@0.5:0.95 of 62.5%. Relative to the YOLOv11n baseline, mAP@0.5 increases by 2.8%, mAP@0.75 rises by 3.7%, mAP@0.5:0.95 gains 2.6%, while recall and F_1-score are improved by 3.3% and 2.5% respectively. Compared with mainstream lightweight detectors, YOLO4SWD outperforms YOLOv12n on precision (82.9% vs. 80.1%), recall (78.1% vs. 73.6%) and mAP@0.5:0.95 (62.5% vs. 58.1%). It also surpasses YOLOv8n by 4.8% on mAP@0.5:0.95 (62.5% vs. 57.7%), with YOLOv10n and YOLOv9n only reaching 53.8% and 43.7% mAP@0.5:0.95 separately.

This paper only evaluates the lightweight performance of the model through GPU-based Params, GFLOPs and FPS indicators, and does not complete practical deployment verification on embedded edge hardware. In addition, the test dataset mainly covers conventional domestic waste, and the detection performance for industrial construction waste needs further exploration in follow-up work. Meanwhile, the model exhibits weak discrimination ability for transparent reflective waste and flexible deformable targets, which restricts its generalization capability in diversified complex scenes.

Follow-up research will focus on three directions. First, the feature extraction branch will be optimized to enhance the differentiation of transparent reflective objects, and an adaptive anchor matching strategy will be designed to improve bounding box localization for flexible targets such as masks. Second, the model will be deployed on embedded hardware including intelligent classification bins and sorting conveyor terminals, to test its inference stability under variable illumination and real-world background conditions. Third, the dataset will be expanded by adding more industrial construction waste and complex transparent target samples, to further boost the generalization performance of the detector for diversified waste identification tasks.

Data availability

Data available on request from the authors.

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Declarations

Competing interests
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